Methane from landfill

After carbon dioxide, methane is the most critical greenhouse gas. Although its atmospheric concentrations are lower than CO_2, methane is 25 times more powerful at trapping heat in the atmosphere. Atmospheric methane comes from various natural sources, including the decay of vegetation in habitats such as bogs and wetlands, but also from livestock rearing, from the use of fossil fuels, and from the decomposition of trash in landfill sites.

In many places, including the UK and US, a number of landfill sites are now trapping and collecting methane to produce energy. Landfill gas contains up to 60 percent methane, depending on the composition of the waste and the age of the site. Vertical and horizontal pipelines are placed through the landfill to collect the methane, which is then processed and filtered. Most of it is used to generate electricity, but it may also be used in industry. After further processing, it can be turned into fuel for vehicles, too.